Breaking the Link: Technical Report on Dataset Description

Data Visualization Project — Deliverable 2

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Table of contents

1 Introduction **2**

2 Dataset Overview **2**
2.1 GHG Emissions (OECD) . . . 2
2.1.1 Structure and Variables . . . 2
2.1.2 Filtering Applied . . . 2
2.1.3 Data Quality . . . 3
2.2 GDP per Capita PPP (World Bank) . . . 3
2.2.1 Structure and Variables . . . 3
2.2.2 Filtering and Transformation . . . 3
2.2.3 Data Quality . . . 3
2.3 PM2.5 Air Pollution Exposure (OECD) . . . 3
2.3.1 Structure and Variables . . . 4
2.3.2 Data Quality . . . 4
2.3.3 Sub-national Region Level (PM2.5 Bubbles Layer) . . . 4
2.3.4 Geocoding of Region Centroids . . . 5
2.4 Climate Projections by SSP Scenario (OECD) . . . 5
2.4.1 Structure and Variables . . . 5
2.4.2 Linking to the Tapio Index . . . 6
2.4.3 Data Quality . . . 6

3 Data Integration and Preprocessing **7**
3.1 Merged Panel . . . 7
3.2 Tapio Elasticity Index . . . 7
3.3 Known Limitations . . . 8

4 Exploratory Data Analysis **8**
4.1 Data Coverage . . . 8
4.2 Tapio Classification Distribution . . . 8
4.3 GDP vs GHG Emissions . . . 9
4.4 PM2.5 Air Quality Trends . . . 10

5 Conclusion **10**

1 Introduction

This technical report describes the datasets used in the *Breaking the Link* data visualization project, which explores the decoupling of economic growth from greenhouse gas (GHG) emissions across countries. The report covers the origin, structure, and quality of each dataset, as well as the preprocessing decisions made to prepare the data for visualization.

The central research question guiding the project is: **Can a nation expand its economy while simultaneously shrinking its environmental footprint?** To answer this, we integrate four primary datasets covering GHG emissions, GDP per capita, air pollution exposure, and climate projections.

2 Dataset Overview

2.1 GHG Emissions (OECD)

Source: OECD Data Explorer — Air and GHG Emissions (`DSD_AIR_GHG@DF_AIR_GHG`)
URL: data-explorer.oecd.org
Format: CSV (long format)
Time range: 1985–2023
Raw rows: ~636,000

2.1.1 Structure and Variables

The dataset contains emissions data across multiple pollutants, measures, and units. The key variables retained after filtering are:

Variable	Description
`REF_AREA`	ISO-3 country code
`TIME_PERIOD`	Year of observation
`OBS_VALUE`	GHG emissions value
`MEASURE`	Type of measure (total, per capita, etc.)
`UNIT_MEASURE`	Unit (tonnes CO_2-equivalent, kg per person, etc.)
`POLLUTANT`	Pollutant type (GHG, CO_2, etc.)

The raw file contains 30 columns, of which 7 are entirely redundant (e.g., structural metadata fields with no analytical value).

2.1.2 Filtering Applied

To obtain a clean, unambiguous time series suitable for the Tapio Index calculation, three filters were applied simultaneously:

- `MEASURE = "_T"` — total emissions, excluding sector breakdowns
- `UNIT_MEASURE = "T_CO2E"` — tonnes of CO_2-equivalent
- `POLLUTANT = "GHG"` — all greenhouse gases (excluding CO_2-only rows which would create duplicate country-year pairs)

These filters reduce the dataset to approximately 2,500 rows covering 94 entities. After excluding aggregate codes (OECD, EU27_2020, OECDA, OECDE, OECDSO), **94 individual countries** remain.

2.1.3 Data Quality

- **Missing values:** Minimal for established OECD members; newer members (e.g., Colombia, Costa Rica) have gaps in early years
- **Encoding issues:** Three country names contain encoding artifacts due to UTF-8 mishandling: `Türkiye`, `China (People's Republic of)`, and `Côte d'Ivoire`. These are corrected during preprocessing via an ISO-3 lookup table
- **Aggregate codes:** Entities such as `OECD`, `OECDA`, `OECDE`, `OECDSO`, and `EU27_2020` are present and must be excluded to avoid inflating country counts
- **Coverage note:** Cuba, Liechtenstein, Monaco, and Vatican appear in the GHG dataset but not in the GDP dataset, resulting in their exclusion from the merged panel

2.2 GDP per Capita PPP (World Bank)

Source: World Bank Open Data — GDP per capita, PPP (current international $)
Indicator: `NY.GDP.PCAP.PP.CD`
URL: data.worldbank.org
Format: CSV (wide format, 4 metadata rows at top)
Time range: 1960–2024

2.2.1 Structure and Variables

Unlike the OECD datasets, the World Bank provides this data in **wide format**, with each year as a separate column header. The file must be melted into long format during preprocessing.

2.2.2 Filtering and Transformation

The dataset is melted from wide to long format, retaining years 1990–2023 to align with the GHG time series. The resulting key variable is `gdp_pc_ppp_usd`.

2.2.3 Data Quality

- **Missing values:** 24 missing values in 2000, 19 in 2020, 21 in 2023 — these occur primarily for smaller or less-reported economies
- **Format challenge:** The 4-row metadata header requires `skiprows=4` when reading with pandas
- **Coverage:** The World Bank dataset covers significantly more countries than the OECD GHG dataset, so the inner join during merging limits the final panel to countries present in both sources

2.3 PM2.5 Air Pollution Exposure (OECD)

Source: OECD Data Explorer — Exposure to Air Pollution (`DSD_AIR_POL@DF_AIR_POLL`)
Format: CSV (long format)
Time range: 1990–2020 (no data available for 2021–2023)
Raw rows: ~244,000

2.3.1 Structure and Variables

The dataset contains 11 rows per country-year combination, corresponding to different measurement types and WHO threshold bands. The key distinction is between:

- `MEASURE = "MEAN_POP"` with `EXPOSURE_LEVEL = "_Z"` — mean population-weighted PM2.5 concentration in μg/m^3 **(selected)**
- `MEASURE = "POP_EXP_POL"` — share of population exposed at 10 different WHO threshold bands (10 rows per country-year) **(excluded)**

We selected the mean concentration measure as it provides an objective, continuous, and internationally comparable value directly aligned with WHO guidelines (5 μg/m^3 annual mean).

Variable	Description
`REF_AREA`	ISO-3 country code
`TIME_PERIOD`	Year of observation
`OBS_VALUE`	Mean PM2.5 concentration (μg/m^3)

2.3.2 Data Quality

- **Time range limitation:** Data ends in 2020. For the dashboard, years 2021–2023 are left as NaN and the UI restricts the PM2.5 view to 2000–2020
- **Duplicate risk:** Without the dual filter (`MEAN_POP` + `_Z`), 11 rows per country-year are returned, causing duplicate key errors and corrupting the Tapio Index calculation (pct_change returns 0 on duplicates, triggering the near-zero GDP guard)
- **Coverage:** Broader than OECD-only, covering most countries in the merged panel

2.3.3 Sub-national Region Level (PM2.5 Bubbles Layer)

The same OECD PM2.5 file also encodes sub-national regional data alongside country aggregates. The distinction is structural: rows where `REF_AREA` equals `ISO` represent the country aggregate, while rows where `REF_AREA` differs from `ISO` represent a sub-national region (e.g. `FR1` for Île-de-France, parent country `FRA`). Previous versions of the pipeline silently dropped these region rows; the current version retains them as a separate output.

Regional bubbles expose intra-country variation invisible at country level — the contrast between heavily industrialised and rural areas — which is directly relevant to the health dimension of the decoupling narrative.

The region-level output (`pm25_regions.csv`) contains **16,629 rows** across **723 sub-national regions** in **52 countries**, covering 1990–2020. It shares the same dual filter (`MEAN_POP` + `_Z`) as the country-level table, and a 5-year rolling mean (`pm25_ugm3_5yr`) is computed at region level using the same `min_periods=3` window applied to the country panel.

Variable	Description
`region_code`	OECD sub-national region code (e.g. `FR1`)
`region_name`	Human-readable region name
`iso3`	Parent country ISO-3 code
`country`	Parent country name

Variable	Description
`year`	Year of observation
`pm25_ugm3`	Mean PM2.5 concentration (μg/m^3)
`pm25_ugm3_5yr`	5-year rolling mean PM2.5 concentration

2.3.4 Geocoding of Region Centroids

Plotly's `Scattergeo` layer requires explicit latitude/longitude coordinates for bubble placement. Since the OECD data provides only region codes and names (no geometry), a separate geocoding step (`geocode.py`) was developed to enrich `pm25_regions.csv` with centroid coordinates.

The geocoder uses the **Nominatim API** (OpenStreetMap) via the `geopy` library, querying each unique region by name and parent country (e.g. `"Île-de-France, France"`). To comply with Nominatim's usage policy, requests are rate-limited to one per 1.2 seconds.

Two categories of lookup failures were encountered and resolved:

1. **OECD naming conventions:** Many region names in the source data include bureaucratic suffixes (`(TL2021)`, `(TL2016)`) or generic English translations (`"North West"`, `"South East"`) that Nominatim cannot resolve because they do not match local administrative names. These were handled via a manually curated override map of 60+ entries mapping OECD labels to locally recognised equivalents (e.g. `"North West"` → `"Severozapaden"` for Bulgaria).
2. **Country name normalisation:** Several OECD country names are incompatible with Nominatim's geocoder (e.g. `"China (People's Republic of)"`, `"Türkiye"`, `"Slovak Republic"`). These are normalised to their common English equivalents before querying.

To avoid losing progress during long runs, the geocoder saves results incrementally after each region lookup, enabling safe resumption if interrupted. The final output (`pm25_regions_geocoded.csv`) achieves **100% coordinate coverage across all 723 regions** — no regions required manual coordinate assignment.

Note on coordinate precision: Nominatim returns the administrative centroid of each region as determined by OpenStreetMap, not a population-weighted centroid. Bubble placement is therefore approximate for regions with uneven population distribution, though the visual effect is acceptable given the scale of a world map.

2.4 Climate Projections by SSP Scenario (OECD)

Source: OECD Data Explorer — Climate Projections (`DSD_REG_CLIM@DF_CLIM_PROJ`)
Format: CSV (~1.5 GB uncompressed)
Time range: 2030–2060
Countries: 52

2.4.1 Structure and Variables

This dataset provides projections of three climate indicators under four SSP scenarios. The pipeline derives two additional columns per metric from the source file: the absolute difference versus the 1981–2010 historical baseline (`_diff_1981_2010`) and the percentage change versus that baseline (`_pct_change_1981_2010`). The dashboard's choropleth map displays the difference columns; the country trend chart displays percentage change:

Variable	Description
`hot_days_proj`	Projected days/year where max temperature > 35°C
`hot_days_proj_diff_1981_2010`	Change vs. 1981–2010 baseline (days/year) — used in map
`trop_nights_proj`	Projected nights/year where min temperature > 20°C
`trop_nights_proj_diff_1981_2010`	Change vs. 1981–2010 baseline (nights/year) — used in map
`icing_days_proj`	Projected days/year where max temperature < 0°C
`icing_days_proj_diff_1981_2010`	Change vs. 1981–2010 baseline (days/year) — used in map

The four scenarios covered are:

Scenario	Label
`PROJ_SSP126`	Low-emissions (SSP1-2.6)
`PROJ_SSP245`	Middle-of-the-road (SSP2-4.5)
`PROJ_SSP370`	High-emissions (SSP3-7.0)
`PROJ_SSP585`	Very high emissions (SSP5-8.5)

2.4.2 Linking to the Tapio Index

A key design decision in this project is linking each country's current Tapio Decoupling classification to a corresponding SSP scenario, creating a narrative bridge between present economic behaviour and projected future climate:

Tapio Classification	**SSP**	**Tapio Classification**	**SSP**
Absolute Decoupling	SSP1-2.6	Expansive Coupling	SSP3-7.0
Relative Decoupling	SSP2-4.5	Recessive Coupling	SSP3-7.0
Recessive Decoupling	SSP2-4.5	Expansive Negative Decoupling	SSP5-8.5
		Strong Negative Decoupling	SSP5-8.5

The linked SSP for each country is determined from its most recent year with a non-undefined Tapio classification, rather than a single global reference year. This avoids leaving out countries whose GDP series ends before the panel's latest year (e.g. China, Colombia, Costa Rica), ensuring every country in the climate dataset receives a linked scenario. A fallback to any available year is applied for countries where all years are classified as undefined. This mapping allows the Future Scenarios view of the dashboard to show users: "if your country continues on its current trajectory, here is what the climate could look like by 2050."

2.4.3 Data Quality

- **File size:** At ~1.5 GB, this file cannot be processed in a single pandas read. It is read in chunks of 100,000 rows and filtered progressively
- **Missing values:** The absolute projection values and difference columns are complete for all 52 countries. The percentage change columns (`_pct_change_1981_2010`) are undefined for countries whose 1981–2010 baseline is near zero (within ± 0.01 days), affecting approximately 24% of countries for hot days and icing days and 2% for tropical nights — typically countries where those weather events are historically absent

- **Coverage limitation:** Only 52 countries are covered, meaning the Future Scenarios view will have incomplete global coverage for countries outside the climate dataset
- **Encoding issues:** Same UTF-8 encoding problems as the GHG dataset for `Türkiye` and `China (People's Republic of)`
- **Column redundancy:** 24 out of 36 columns are irrelevant metadata and are dropped during preprocessing

3 Data Integration and Preprocessing

3.1 Merged Panel

The preprocessing pipeline (`panel.py`) produces two output files from the four source datasets.

The primary output, `merged_panel.csv`, is built by joining datasets using `(iso3, year)` as a composite key:

1. **GHG total emissions** ← inner join with **GDP per capita** (only countries with both variables)
2. Result ← left join with **GHG per capita** (tooltip layer)
3. Result ← left join with **PM2.5 country-level** (2021–2023 will be NaN by design)

The final panel covers **90 countries** and **34 years** (1990–2023), yielding **2,420 rows**.

A second output, `pm25_regions.csv`, is produced independently from the sub-national branch of the PM2.5 source file (see Section 2.3). It is not merged into the main panel — instead it is loaded separately by the dashboard and enriched with geographic coordinates via the geocoding step described above, producing `pm25_regions_geocoded.csv` as the final input to the PM2.5 bubble layer.

3.2 Tapio Elasticity Index

The Tapio Decoupling Elasticity Index (E) is computed as:

$$E = \frac{\Delta\%\text{GHG}}{\Delta\%\text{GDP}}$$

Where $\Delta\%$ denotes the year-over-year percentage change. The index is left undefined when GDP growth is near zero ($|\Delta\%\text{GDP}| < 0.01$) to avoid division by extremely small numbers.

A **5-year rolling mean** of E is also computed to smooth outliers (like COVID-2020) and other short-term shocks — providing a more stable signal for the choropleth map. The classification follows the typology introduced by Tapio (2005) and applied in subsequent decoupling studies, including Wang et al. (2022), which informed the threshold values and category boundaries used in this project. Countries are classified into eight categories:

Category	Condition	Category	Condition
Absolute Decoupling	GDP↑, $E \leq 0$	Recessive Decoupling	GDP↓, $E > 1.2$
Relative Decoupling	GDP↑, $0 < E < 0.8$	Recessive Coupling	GDP↓, $0 < E \leq 1.2$
Expansive Coupling	GDP↑, $0.8 \leq E < 1.2$	Strong Negative Decoupling	GDP↓, $E \leq 0$
Expansive Negative Decoupling	GDP↑, $E \geq 1.2$	Undefined	Near-zero GDP growth

3.3 Known Limitations

- **PM2.5 gap (2021–2023):** No PM2.5 data is available for recent years. The dashboard restricts the air quality view to 2000–2020 and notes this limitation explicitly
- **Country coverage asymmetry:** The climate projections dataset covers 52 countries, while the historical panel covers 90. This means the Future Scenarios view will have incomplete global coverage for countries outside the climate dataset
- **GHG per capita vs total:** The Tapio Index is calculated using **total** GHG emissions (not per capita), as the decoupling relationship is between aggregate economic output and aggregate environmental pressure. Per capita figures are retained as a tooltip layer only
- **Interpolation:** Linear interpolation is applied within country groups to fill internal gaps in the PM2.5 and GDP series, which may introduce slight inaccuracies for countries with extended missing periods
- **Region centroid approximation:** Sub-national PM2.5 bubble positions are based on Nominatim administrative centroids rather than population-weighted centroids. For large or unevenly populated regions, the bubble may appear displaced relative to where most of the exposed population lives
- **Geocoding reproducibility:** Nominatim results depend on OpenStreetMap's current data and may change between runs as OSM coverage evolves. The geocoded coordinates are therefore committed to the repository as a static file rather than re-generated at runtime

4 Exploratory Data Analysis

4.1 Data Coverage

Figure 1 shows the percentage of countries with available data for each key variable across years. GHG, GDP, and GHG per capita are complete throughout the panel. PM2.5 stands out as the most significant gap — data is entirely missing for 2021–2023 and sparsely available before 2000, which directly shapes the design of the Air Quality layer in the dashboard.

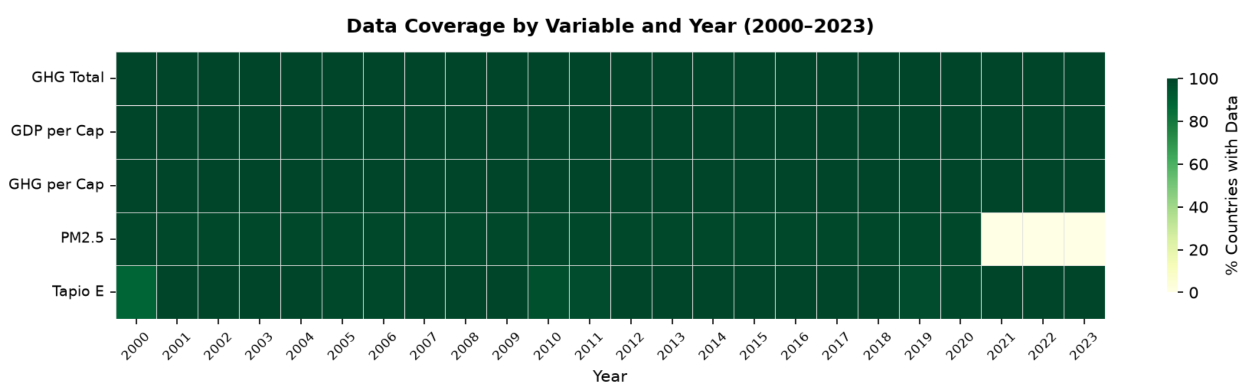

Figure 1: Data coverage heatmap across variables and years.

4.2 Tapio Classification Distribution

Figure 2 shows how countries are distributed across Tapio categories each year. The growing dominance of absolute and relative decoupling from 2010 onwards reflects the global shift toward cleaner energy. The COVID-19 spike in 2020 is clearly visible — many countries temporarily enter recessive categories due to simultaneous GDP and emissions collapses, confirming the need for the 5-year rolling average in the dashboard.

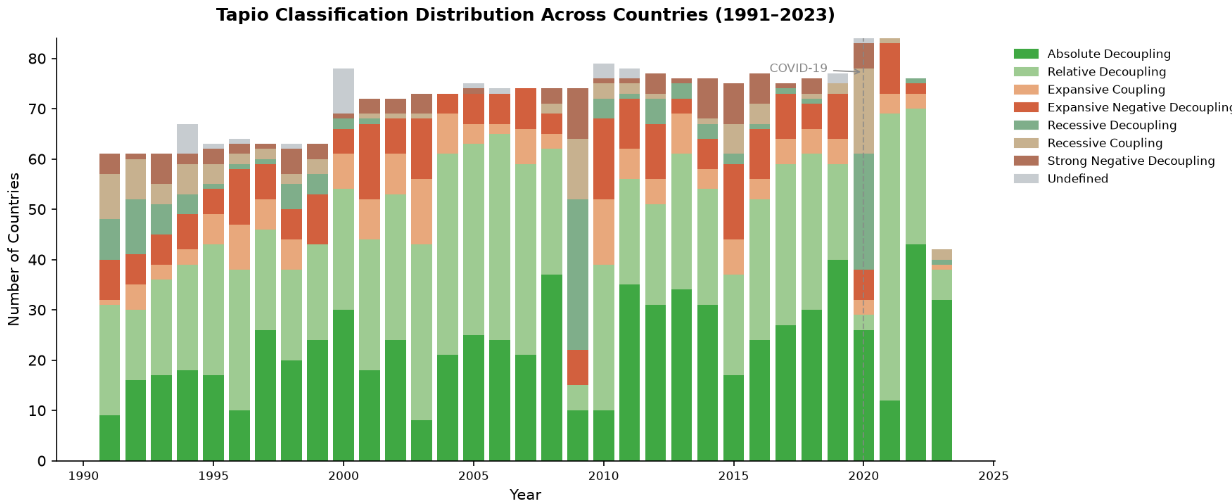

Figure 2: Stacked bar chart of Tapio classifications across all countries, 1991–2023.

4.3 GDP vs GHG Emissions

Figure 3 compares GDP per capita against GHG emissions per capita in 2000 and 2023. The shift is striking — in 2023, countries are richer but clustered at lower emission levels, and the dominant colour has shifted toward dark green (absolute decoupling). This directly validates the core narrative of the dashboard.

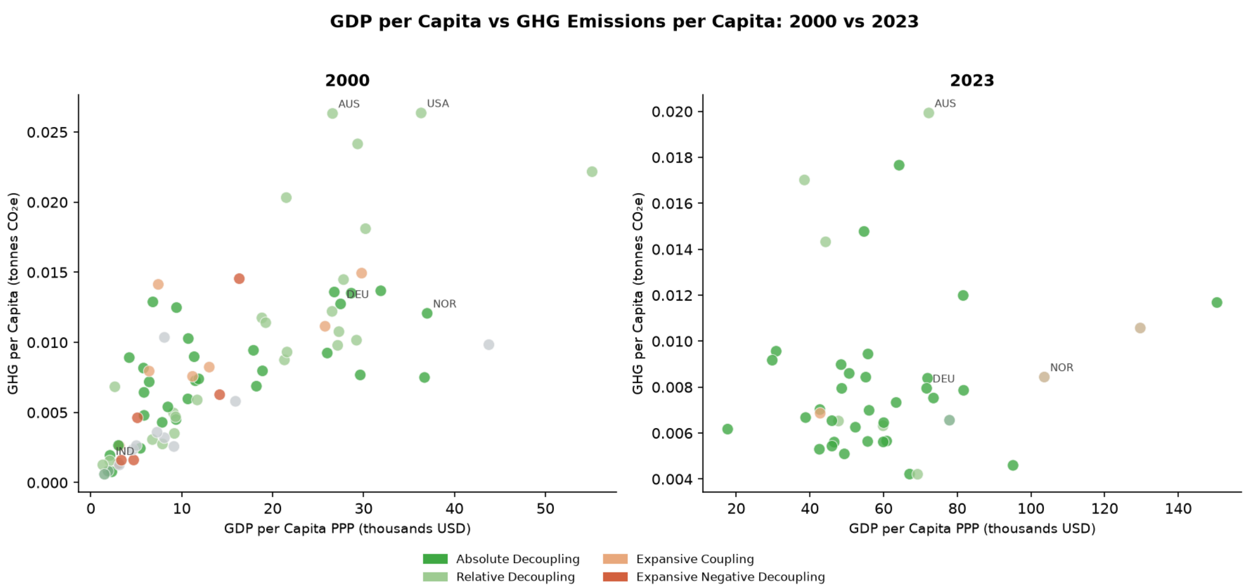

Figure 3: Scatter plot of GDP per capita vs GHG per capita in 2000 and 2023, coloured by Tapio classification.

4.4 PM2.5 Air Quality Trends

Figure 4 shows PM2.5 exposure trends for a selection of countries. Western nations (Germany, USA, Japan) show consistent decline toward but still above the WHO guideline of 5 μg/m^3. China shows a declining trend after peaking around 2006. India remains persistently high. The data gap after 2020 is clearly marked and is acknowledged as a limitation in the Air Quality layer of the dashboard.

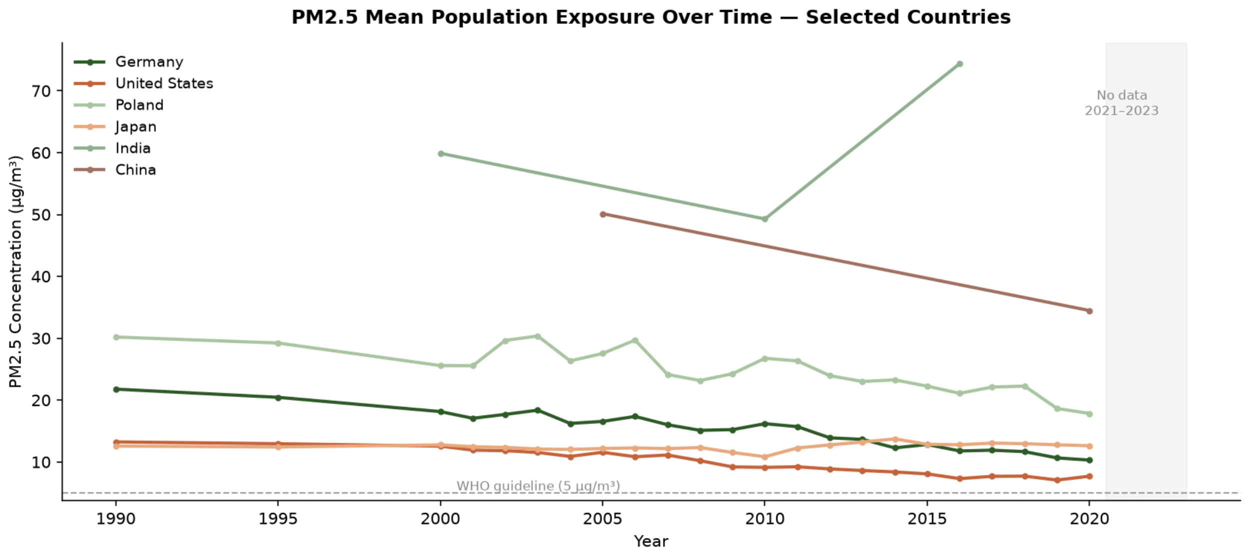

Figure 4: PM2.5 mean population exposure over time for selected countries.

5 Conclusion

The four datasets used in this project are of generally high quality and well-suited for the intended visualization. The primary challenges encountered were encoding inconsistencies in country names, the 11-row-per-country-year structure of the PM2.5 dataset requiring careful filtering, and the large file size of the climate projections dataset requiring chunk-based processing. All issues have been addressed in the preprocessing pipeline (`panel.py`), which is fully reproducible and documented in the project repository.

The merged panel of 90 countries across 34 years provides sufficient coverage to tell a compelling global story about economic decoupling, with the climate projections adding a forward-looking narrative dimension that connects present behaviour to future consequences.

Report prepared as Deliverable 2 for the Data Visualization course.
Data sources: OECD Data Explorer, World Bank Open Data.
Code repository: github.com/kristine-p/economic-decoupling-dataviz
Methodological reference: Tapio, P. (2005). Towards a theory of decoupling. Transport Policy, 12(2), 137–151.
Applied framework: Wang et al. (2022). Carbon emissions and economic growth in the Yellow River Basin.
researchgate.net/publication/365861873